Leader-Follower Tracking with Collision Avoidance for Omni-directional Mobile Robots: Linear vs Homogeneous Controller

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Abstract: This paper deals with the leader-follower tracking control for omni-directional mobile robots, the controller of the leader omni-directional mobile robot is unknown and the safe region is given. A linear controller is firstly designed for the follower omni-directional mobile robot such that the asymptotical stability is guaranteed. Based on the linear controller, a homogeneous controller is obtained, and the error system is guaranteed to be stabilized in finite-time. Compared with the linear controller, the homogeneous controller has the properties of faster response and better robustness. Furthermore, the convergence rate of the homogeneous controller can be simply tuned by properly selecting a homogeneity degree. The simulation results are given to illustrate the effectiveness of the designed controllers.

Key Words: Homogeneous System, Leader-follower Tracking, Omni-directional Mobile Robot, Finite-Time Control

1 Introduction

The common wheeled mobile robots are unable to achieve holonomic drive and allows for simultaneous translation and rotation movements. However, omni-directional mobile robots are a category of special wheeled mobile robots that have the capability to move in any direction on a flat surface [1]. Omni-directional mobile robots have been widely applied in industrial, educational, medical and other fields [2–4]. Over the past few decades, several control algorithms have been designed for the control of omni-directional mobile robots, such as adaptive control [5], sliding-mode control [6], model predictive control [7]. In [5], an adaptive control method based on a dynamic model of omni-directional mobile robots is presented. The method in [5] is able to adjust parameters to optimize the control performance of the system, but it relies on a dynamic model of the system, which may not always be available. In [6], a fuzzy terminal sliding-mode control for omni-directional mobile robots is proposed, but the method can suffer from the chattering problem, which leads to a degradation in control accuracy. In [7], a model predictive control approach bases on the kinematics model of the omni-directional mobile robots and considers the dynamic constraints, but it requires substantial computational resources.

There have been growing interests in finite-time control of omni-directional mobile robots due to its ability to achieve rapid convergence and better robustness. Some finite-time control algorithms have been developed for omni-directional mobile robots, such as [8, 9]. In [8], a finite-time convergent interval type-2 fuzzy sliding-mode controller of omni-directional mobile robots with system uncertainties and external disturbances is designed. In [9], a new task space nonsingular terminal sliding mode manifold to control omni-directional mobile robots is introduced. The approaches in [8, 9] are often plagued by high computational complexity and difficulties in adapting to system dynamics changes. Recently, some finite-time controllers have been developed based on homogeneity, where the finite-time convergent rate of the homogeneous controller can be tuned by a proper selection of the homogeneity degree. The properties of homogeneous control are described as follows: symmetry pertains to the property of an object where certain characteristics remain unchanged under specific transformations. Homogeneity, on the other hand, is a form of dilation symmetry that is applicable to both linear and many nonlinear models in mathematical physics [10–12]. Homogeneous models serve as local approximations for control systems when linearization is overly conservative, lacks information, or simply cannot be applied [13]. Compared to linear systems, homogeneous systems have several notable properties including faster convergence, better robustness, and reduced overshoots. An important characteristic of homogeneous systems is that their convergent rate is determined by the degree of homogeneity.

A critical performance metric for leader-follower tracking control problem is the control precision. Classic linear control methods often result in lower control accuracy within asymptotical convergent rate. However, homogeneous control can achieve high control accuracy in finite-time convergence. The purpose of this paper is to develop a homogeneous control approach based on generalized homogenization of linear system to deal with the leader-follower tracking control problem for two omni-directional mobile robots. The design of the homogeneous control method bases on the canonical homogeneous norm in [14, 15], and has simple parameter tuning rules. Compared with the linear controller with asymptotical convergence, the finite-time stabilization of leader–follower tracking for two omni-directional mobile robots with a safe zone by using generalized homogeneous controllers is realized. Moreover, the homogeneous controller can make the follower omni-directional mobile robot fast track the leader omni-directional mobile robot with disturbances.

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Notation: Define $\mathbb{R}$ as the set of real numbers, $\mathbb{R}_+$ as the set of non-negative real numbers. The exponential function with the base of the natural constant is represented as $\exp(\cdot)$; $\|\cdot\|$ in $\mathbb{R}^n$ is denoted as a norm, $\mathbf{0}$ represents the zero vector in $\mathbb{R}^n$. The canonical Euclidean norm of $x \in \mathbb{R}^n$ is represented as $\|x\|_2 = \sqrt{x^{\mathrm{T}}x}$. The notation $P > 0(< 0, \leq 0, \geq 0)$ for $P \in \mathbb{R}^{n\times n}$ indicates that the matrix P is symmetric and positive (negative) definite (semidefinite). The term L^∞ is the set of the essentially bounded measurable functions. C^1 is the set of continuously differentiable. $\mathbf{Co}\{\cdot\}$ means that it is a convex combination of the elements in $\{\cdot\}$. The diagonal matrix denoted as $\mathrm{diag}\{\Lambda_1, \Lambda_2\}$ consists of two matrices Λ_1 and Λ_2.

2 Problem statement

Consider the following dynamic model of the leader and follower omni-directional mobile robots:

$$\begin{aligned} m\ddot{r}_1 + Q_1(r_1, \dot{r}_1) = u_1 \\ m\ddot{r}_2 + Q_2(r_2, \dot{r}_2) = u_2 \end{aligned} \tag{1}$$

where $r_1 = [x_1, y_1]^{\mathrm{T}} \in \mathbb{R}^2$ is the coordinate of the centroid for the leader omni-directional mobile robot, and $r_2 = [x_2, y_2]^{\mathrm{T}} \in \mathbb{R}^2$ is the coordinate of centroid for the follower omni-directional mobile robot; m is the mass of omni-directional mobile robots (assumed to be known), and Q_1, Q_2 are generalized forces (assumed to be unknown). The control input $u_1 = [u_{1x}, u_{1y}]^{\mathrm{T}} \in \mathbb{R}^2$ of the leader omni-directional mobile robot is bounded but unknown, and the control input $u_2 = [u_{2x}, u_{2y}]^{\mathrm{T}} \in \mathbb{R}^2$ is required to design for the follower omni-directional mobile robot to track the leader omni-directional mobile robot. In order to avoid collisions, a safety zone $\mathbf{D}_{safe}$ around the centroid r_1 for the leader omni-directional mobile robot is designed as follows:

$$\mathbf{D}_{safe} = \mathbf{Co}\{d_1, d_2, ..., d_n\} \tag{2}$$

where $\mathbf{D}_{safe}$ is a convex set consisted of n security points, $d_i = -[cos(2\pi i/n), sin(2\pi i/n)]^{\mathrm{T}} \in \mathbb{R}^2, i = 1, 2, ..., n$, are uniform distributed on a circle with the centroid r_1 for the leader and the radius r. The vectors e defined in (3) denote the tracking error, i.e.,

$$e = r_2 - r_1 - d_{i^*} \tag{3}$$

where the index $i^* = min(argmin(\|r_2 - r_1 - d_i\|))$ corresponds to the security point being closest to r_2. For example, in Fig.1, the red robot indicates follower omni-directional mobile robot and the blue robot indicates leader omni-directional mobile robot. The centroids of two omni-directional mobile robots are r_1 and r_2, respectively. $\mathbf{D}_{safe}$ is a square area consisting of four security points, d_1, d_2, d_3 and d_4. The tracking error is $e = r_2 - r_1 - d_1$ with $i^* = 1$. Then, the control input u_2 for follower omni-directional mobile robot is designed as:

$$u_2 = K_1 e + K_2 \dot{e} \tag{4}$$

where K_1 and K_2 are two-dimensional diagonal matrices. For a time-invariant i^*, the error equation is represented as

$$m\ddot{e} = u + \gamma \tag{5}$$

where $u = [u_x, u_y]^{\mathrm{T}} = u_2 = K_1 e + K_2 \dot{e}$ and $\gamma = Q_1(r_1, \dot{r}_1) - u_1 - Q_2(r_2, \dot{r}_2)$ are unknown components of the control system treated as a disturbance and $\gamma \in L^\infty$. Define the system state $x = [e, \dot{e}]^{\mathrm{T}} = [e_x, e_y, \dot{e}_x, \dot{e}_y]^{\mathrm{T}}$, where e_x and e_y denote the components of e in x and y coordinates, and $\dot{e}_x$ and $\dot{e}_y$ denote the components of $\dot{e}$ in x and y coordinates, respectively. Then system (5) can be represented as the following state space model:

$$\dot{x} = Ax + B(u + \gamma) \tag{6}$$
$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & \frac{1}{m} \end{bmatrix}$$

and the pair $\{A, B\}$ is controllable.

The purpose of the control is to design u such that the follower omni-directional mobile robot can track the leader omni-directional mobile robot in finite-time. First we design linear stabilizing controller and avoid collisions using the technique suggested in Section 4. In Section 5, the linear stabilizing controller is upgraded to homogeneous stabilizing controller with the consideration of avoid collisions.

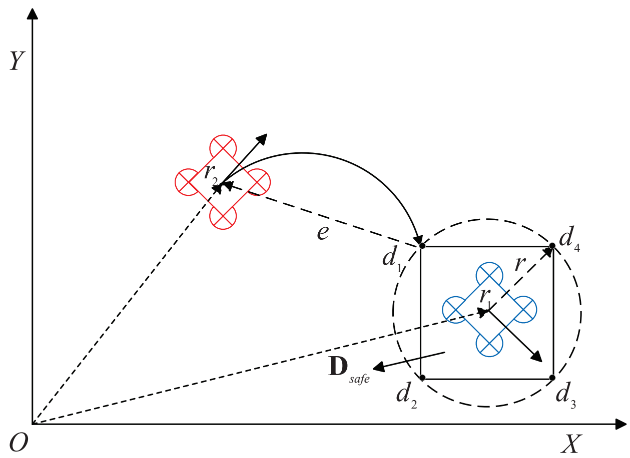

Fig. 1: The illustration of the leader-follower tracking control for two omni-directional mobile robots

3 Preliminaries

The homogeneity (dilation symmetry) based analysis and design of the control system require a dilation to be specified. By definition, a dilation $\mathbf{d}(s)$, where $s \in \mathbb{R}$, refers to a one-parameter group of transformations that satisfies the limit property $\lim\limits_{s\to+\infty} \|\mathbf{d}(s)x\| = +\infty$ and $\lim\limits_{s\to-\infty} \|\mathbf{d}(s)x\| = 0$, for all $x \neq 0$. The linear dilation in $\mathbb{R}^n$ is defined as in [15], denoted as $\mathbf{d}(s) = \exp(sG_{\mathbf{d}}) = \sum_{i=0}^{+\infty} \frac{s^i G^i_{\mathbf{d}}}{i!}$, $s \in \mathbb{R}$, where $G_{\mathbf{d}} \in \mathbb{R}^{n\times n}$ is an anti-Hurwitz matrix referred to as a generator of $\mathbf{d}$.

Definition 1. *[15]. The canonical $\mathbf{d}$-homogeneous norm in $\mathbb{R}^n$ is denoted by $\|\cdot\|_{\mathbf{d}} : \mathbb{R}^n \to \mathbb{R}_+$ and is defined as $\|0\|_{\mathbf{d}} = 0$ and $\|x\|_{\mathbf{d}} = \exp(s_x)$, where $s_x \in \mathbb{R} : \|\mathbf{d}(-s_x)x\| = \|\mathbf{d}(-\ln\|x\|_{\mathbf{d}})x\| = 1$ for $x \neq 0$. Here, $\mathbf{d}$ represents a linear monotone dilation in $\mathbb{R}^n$.*

Definition 2. *[16]. A vector field $f : \mathbb{R}^n \to \mathbb{R}^n$ is defined as $\mathbf{d}$-homogeneous of a degree $\mu \in \mathbb{R}$ if $f(\mathbf{d}(s)x) =$*

$\exp(\mu s)\mathbf{d}(s)f(x)$*, holds for all* $x \in \mathbb{R}^n$ *and* $s \in \mathbb{R}$*, where* $\mathbf{d}$ *is a linear dilation in* $\mathbb{R}^n$*.*

The canonical homogeneous norm is derived from a conventional norm in $\mathbb{R}^n$. It is important to note that if $\|\cdot\| \in C^1(\mathbb{R}^n\backslash\{\mathbf{0}\})$ then $\|\cdot\|_{\mathbf{d}} \in C^1(\mathbb{R}^n\backslash\{\mathbf{0}\})$. Therefore, it can be concluded [15] that

$$\frac{\partial\|x\|_{\mathbf{d}}}{\partial x} = \|x\|_{\mathbf{d}} \frac{x^{\mathrm{T}}\mathbf{d}^{\mathrm{T}}(-\ln\|x\|_{\mathbf{d}})P\mathbf{d}(-\ln\|x\|_{\mathbf{d}})}{x^{\mathrm{T}}\mathbf{d}^{\mathrm{T}}(-\ln\|x\|_{\mathbf{d}})PG_{\mathbf{d}}\mathbf{d}(-\ln\|x\|_{\mathbf{d}})x} \quad (7)$$

for $x \neq 0$, assuming that $\|x\|_{\mathbf{d}}$ represents the canonical homogeneous norm that is induced by the norm $\|x\| = \sqrt{x^{\mathrm{T}}Px}$ with $P = P^{\mathrm{T}} \in \mathbb{R}^{n\times n}$ satisfying the conditions $P > 0$ and $PG_{\mathbf{d}} + G_{\mathbf{d}}^{\mathrm{T}}P > 0$.

4 Linear tracking control

According to (5) in the disturbance-free case ($\gamma = 0$), an ordinary differential equation is derived

$$m\ddot{e} = K_1 e + K_2\dot{e} \quad (8)$$

where $e(0) > 0$. According to the stability analysis of the system, if $K_1 < 0$ and $K_2 < 0$, then the system (8) is stable and $e(t) \to 0$ as $t \to \infty$. However, there is no guarantee that $e(t) \geq 0, \forall t \geq 0$. It means that there may exist some overshoots that make tracking error $e(t) < 0$. In this case, it is possible that follower omni-directional mobile robot enter into the sage region of leader omni-directional mobile robot, and the collision happens. The following Lemma 1, inspired by [17], guarantees that the tracking errors $e(t) \geq 0, \forall t \geq 0$, and $e(t)$ is asymptotical convergent to 0 when $t \to \infty$.

Lemma 1. *Suppose that* K_1 *and* K_2 *satisfy the following conditions*

$$K_1 = \frac{\lambda I(K_2 + \lambda I)}{m} < 0, K_2 < -\lambda I < 0 \quad (9)$$

where λ *is a positive scalar. Then, for* $\lambda \geq \max\{\frac{-m\dot{e}_x(0)}{e_x(0)}, \frac{-m\dot{e}_y(0)}{e_y(0)}\}$*, the solution of the tracking errors in* (8) *satisfy* $e(t) \geq 0, \forall t \geq 0$*, and* $e(t) \to 0$ *as* $t \to \infty$*.*

Proof. Introducing the new variable

$$g = m\dot{e} + \lambda I e \quad (10)$$

By utilizing (10), the derivative of e is derived as

$$\dot{e} = \frac{-\lambda I e + g}{m} \quad (11)$$

The derivative of g with respect to t is

$$\begin{aligned}\dot{g} &= m\ddot{e} + \lambda I\dot{e} = K_1 e + K_2\dot{e} + \lambda I\dot{e} \\ &= (K_1 - \frac{\lambda I(K_2 + \lambda I)}{m})e + \frac{K_2 + \lambda I}{m}g \end{aligned} \quad (12)$$

By letting $K_1 - \frac{\lambda I(K_2+\lambda I)}{m} = 0$ and $\frac{K_2+\lambda I}{m} < 0$, and solving the ordinary differential equations (11) and (12), it is obtained that

$$\begin{aligned} e(t) &= \exp(-\frac{\lambda I}{m}t)e(0) + \int_0^t \exp(-\frac{\lambda I}{m}(t-s))\frac{g(s)}{m}\,ds \\ g(t) &= \exp(\frac{K_2 + \lambda I}{m}t)g(0) \end{aligned} \quad (13)$$

According to (10) and (13), if $\lambda I \geq \mathrm{diag}\{\frac{-m\dot{e}_x(0)}{e_x(0)}, \frac{-m\dot{e}_y(0)}{e_y(0)}\}$, then $g(t) \geq 0$, $e(t) \geq 0, t \geq 0$. Therefore, when the matrices K_1, K_2 and λI satisfy the following conditions:

$$\begin{aligned} &\lambda I \geq \mathrm{diag}\{\frac{-m\dot{e}_x(0)}{e_x(0)}, \frac{-m\dot{e}_y(0)}{e_y(0)}\}, t \geq 0 \\ &K_1 = \frac{\lambda I(K_2 + \lambda I)}{m} < 0, K_2 < -\lambda I < 0 \end{aligned} \quad (14)$$

the tracking error $e(t) \geq 0, \forall t \geq 0$, and $e(t) \to 0$ as $t \to \infty$. In order to facilitate the subsequent simulation, K_1 and K_2 are represented as

$$K_1 = \frac{\lambda I(K_2 + \lambda I)}{m}, K_2 = -\sigma\lambda I \quad (15)$$

where $\sigma > 1$ is a given scalar. The control input u based on the linear feedback controller is

$$u = K_{lin}x, K_{lin} = [K_1, K_2] \quad (16)$$

According to Lemma 1, the unperturbed system (8) with the linear feedback controller u can guarantee that $e(t) \geq 0, \forall t \geq 0$, and $e(t) \to 0$ as $t \to \infty$ and has asymptotic convergent rate. However, for system (5), the linear feedback controller u cannot always guarantee that $e(t) \geq 0, \forall t \geq 0$, and the tracking errors are convergent to the neighborhood of 0 due to bounded disturbance γ.

5 Homogeneous tracking control

Since the linear feedback controller is not able to satisfy finite-time control, to achieve finite-time control, the linear feedback controller (16) can be upgraded to a homogeneous feedback controller. By utilizing Definitions 1 and [12], the homogeneous feedback controller is

$$u = K_0 x + u_{hom} \quad (17)$$

where the homogeneous stabilizing feedback part u_{hom} is

$$u_{hom} = \|x\|_{\mathbf{d}}^{1+\mu}(K_{lin} - K_0)\mathbf{d}(-\ln\|x\|_{\mathbf{d}})x \quad (18)$$

the scalar μ is the homogeneity degree; $K_0 = Y_0(G_0 - I)^{-1}$. Y_0 and G_0 are solutions of the linear algebraic equation $AG_0 - G_0 A + BY_0 = A$ where $G_0 B = \mathbf{0}$. $\|x\|_{\mathbf{d}}$ is the canonical homogeneous norm induced by the norm $\|x\| = \sqrt{x^{\mathrm{T}}Px}$, where $P > 0, PG_{\mathbf{d}} + G_{\mathbf{d}}^{\mathrm{T}}P > 0$, $P(A + BK_{lin}) + (A + BK_{lin})^T P < 0$ and $G_{\mathbf{d}} = I + \mu G_0$. Let $\rho > 0$ be such that we denote $P(A + BK_{lin}) + (A + BK_{lin})^{\top}P + \rho(PG_{\mathbf{d}} + G_{\mathbf{d}}^{\top}P) \leq 0$. By substituting (17) into (6) and according to Definition 2, the system (6) is $\mathbf{d}$-homogeneous of a degree μ. Inspired by [14] we prove the following theorem.

Theorem 1. *If* $(1 - m^2 + \sigma(m-1))\lambda^2 + \rho m^2\lambda \geq 0$ *then, under condition of Lemma 1, the system* (6) *with the homogeneous controller* (17) *is finite-time stable, nonovershooting without disturbance (i.e.,* $e(t) \geq 0, \forall t \geq 0$*) and input-to-state stable with disturbance.*

Proof. Denote the system (6) as $\dot{x} = f(x, \gamma)$, and its disturbance-free system as $\dot{x} = f(x, 0)$. Let the canonical homogeneous norm $\|x\|_{\mathbf{d}}$ be a Lyapunov function candidate for the system. The time derivative of the canonical homogeneous norm for the disturbance-free system is given by (7), hence

$$\frac{d\|x\|_{\mathbf{d}}}{dt} = \frac{\partial \|x\|_{\mathbf{d}}}{\partial x} f(x,0) \tag{19}$$

Let $s = \ln \|x\|_{\mathbf{d}}$, it is obtained that

$$\frac{\partial \|x\|_{\mathbf{d}}}{\partial x} f(x,0) = \|x\|_{\mathbf{d}} \frac{(\mathbf{d}_x(-s)x)^{\mathrm{T}} P \mathbf{d}_x(-s) f(x,0)}{(\mathbf{d}_x(-s)x)^{\mathrm{T}}(PG_{\mathbf{d}})\mathbf{d}_x(-s)x} \tag{20}$$

Let $\mathbf{d}_x(-s)x = z$, then (20) is represented as

$$\frac{\partial \|x\|_{\mathbf{d}}}{\partial x} f(x,0) = \|x\|_{\mathbf{d}} \frac{z^{\mathrm{T}} P \mathbf{d}(-s)(Ax+Bu)}{z^{\mathrm{T}} P G_{\mathbf{d}} z} \tag{21}$$

By utilizing (17) and (18), (21) is derived as

$$\begin{aligned} &\frac{d\|x\|_{\mathbf{d}}}{dt} = \frac{\partial \|x\|_{\mathbf{d}}}{\partial x} f(x,0) \\ &\leq \|x\|_{\mathbf{d}}^{1+\mu} \frac{z^{\mathrm{T}} P^{-1} \{(A_0+BK)^{\mathrm{T}} P + P(A_0+BK)\} P^{-1} z}{z^{\mathrm{T}} P^{-1} \{G_{\mathbf{d}} P + P G_{\mathbf{d}}^{\mathrm{T}}\} P^{-1} z} \\ &= -\rho \|x\|_{\mathbf{d}}^{1+\mu} < 0 \end{aligned} \tag{22}$$

where $A_0 = A + BK_0$ and $K = K_{lin} - K_0$. Therefore, (22) indicates that the unperturbed system (6) is asymptotically stable. Let $V(x) = \|x\|_{\mathbf{d}}$ be a $\mathbf{d}$-homogeneous Lyapunov function. In this case, (22) is equivalent to

$$\frac{1}{-\mu}(V^{-\mu}(x(t,x_0)) - V^{-\mu}(x_0)) \leq -\rho t \tag{23}$$

where $\forall x_0 \in \mathbb{R}^n$. If $\mu < 0$ then $V(x(t,x_0)) = 0, \forall t \geq \frac{V^{-\mu}(x_0)}{-\mu\rho}$. According to the definition of finite-time stability in [18], the unperturbed system (6) is finite-time stable.

To guarantee that the unperturbed system (6) is nonovershooting, i.e., the tracking errors $e(t) \geq 0, \forall t \geq 0$, consider a vector-valued homogeneous barrier function ϕ defined as follows $\phi(x) = H\mathbf{d}(-\ln \|x\|_{\mathbf{d}})x$, where $H = (h_1, h_2)^{\mathrm{T}}$, $h_i = \ell_1^{\mathrm{T}}(A + \lambda I)^{i-1}$ and $\lambda > 0$, $i = 1, 2$; $\ell_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}^{\mathrm{T}}$. Then we have

$$\begin{aligned} \dot{\phi} = &-H \frac{\frac{d\|x\|_{\mathbf{d}}}{dt}}{\|x\|_{\mathbf{d}}} G_{\mathbf{d}} \mathbf{d}(-\ln \|x\|_{\mathbf{d}})x \\ &+ H\mathbf{d}(-\ln \|x\|_{\mathbf{d}})(A_0 x + BK\|x\|_{\mathbf{d}}^{1+\mu} \mathbf{d}(-\ln \|x\|_{\mathbf{d}})x) \end{aligned} \tag{24}$$

By denoting $\alpha = -\|x\|_{\mathbf{d}}^{-1-\mu} \frac{d\|x\|_{\mathbf{d}}}{dt} \geq \rho$ and using the identities $\mathbf{d}(s)A_0 = \exp(s)\mathbf{d}(s)A_0$, $\mathbf{d}(s)B = \exp(s)B$ we derive

$$\dot{\phi} = \|x\|_{\mathbf{d}}^{\mu} \Pi \phi \tag{25}$$

where $\Pi = \begin{bmatrix} -\lambda + 2\alpha & 0 & 1 & 0 \\ 0 & -\lambda + 2\alpha & 0 & 1 \\ M & 0 & N & 0 \\ 0 & M & 0 & N \end{bmatrix}$, $M = \frac{(1-m^2+\sigma(m-1))\lambda^2}{m^2} + \lambda\alpha$ and $N = (\frac{-\sigma}{m} + 1)\lambda + \alpha$. Since $(1 - m^2 + \sigma(m-1))\lambda^2 + \rho m^2 \lambda \geq 0$ then Π is a Metzler matrix and the system (25) is positive. Therefore, there exists a homogeneous cone $\Omega = \{x : h_i \mathbf{d}(-\ln \|x\|_{\mathbf{d}})x \geq 0, i = 1, 2\}$, which is a strictly positively invariant set of the unperturbed system (6) and guarantees $e(t) \geq 0, \forall t \geq 0$.

In order to prove that the system (6) is input-to-state stable, define $V(x)$, s and z as the same way in above. Taking $\mathbf{d}_\gamma(s) = \exp((1+\mu)s)$ we derive

$$\begin{aligned} \dot{V}(x) &= \frac{\partial V}{\partial x} f(x,\gamma) = \frac{\partial V}{\partial x} f(x,0) + \frac{\partial V}{\partial x} f(x,\gamma) - f(x,0) \\ &\leq -\rho V^{1+\mu} + V^{1+\mu} \frac{\partial V(z)}{\partial z}(f(z, \mathbf{d}_\gamma(-s)\gamma) - f(z,0)) \end{aligned} \tag{26}$$

where $z = \mathbf{d}_x(-\ln V)x$ belongs to a compact, so $\exists k > 0 : \left\| \frac{\partial V(z)}{\partial z} \right\| \leq k$. Since $f(x,\gamma)$ is continuous, then $\forall C > 0, \exists \omega_c \in \mathcal{K}_\infty$ such that

$$\|\mathbf{d}_\gamma(-s)\gamma\| \leq C, \|f(z, \mathbf{d}_\gamma(-s)\gamma) - f(z,0)\| \leq \omega_c(\|\mathbf{d}_\gamma(-s)\gamma\|) \tag{27}$$

Therefore, for $C = \omega_c^{-1}(0.5\rho/k)$, it can be derived as

$$\dot{V} \leq -0.5\rho V^{1+\mu}, \|\frac{\gamma}{C}\|_{\mathbf{d}_\gamma} \leq V(x) \tag{28}$$

Due to $\gamma \in L^\infty$ and $\bar{\gamma}(t) = \sup_{\tau \in (0,t)} \|\gamma(\tau)/C\|_{\mathbf{d}}$, then

$$V(x(t)) \leq \max\{\beta(V(x_0),t), \bar{\gamma}(t)\} \leq \beta(V(x_0),t) + \bar{\gamma}(t) \tag{29}$$

where $\beta(V(x_0),t) = \exp(-0.5\rho)V(x_0)$ and $\forall t \geq 0$. According to the definition of input-to-state stabilization in [19], it is obtained that the system (6) is input-to-state stable. The proof is complete.

Homogeneous feedback controller (17) may result in a large control input, to deal with this problem, the upgrade of (17) is

$$u = K_0 x + u'_{hom} \tag{30}$$

where the homogeneous stabilizing feedback part u'_{hom} is

$$u'_{hom} = \mathrm{sat}_{a,b}^{1+\mu}(\|x\|_{\mathbf{d}})(K_{lin} - K_0)\mathbf{d}(-\ln \mathrm{sat}_{a,b}(\|x\|_{\mathbf{d}}))x \tag{31}$$

where $\mathrm{sat}_{a,b}(\delta) = \min\{\max\{a,\delta\}, b\}$ where $\delta > 0$ satisfy $(A + BK_{lin})^{\mathrm{T}} P + P(A + BK_{lin}) + 2\delta P < 0$ for all $a, b \in \mathbb{R}$. The implementation of the linear controller and homogeneous controller for the leader-follower tracking control of two omni-directional mobile robots are summarized in Algorithm 1.

6 Simulation Example

In this section, a simulation example is provided to demonstrate the designed linear and homogeneous controllers. In the simulation, the safe zone $\mathbf{D}_{safe}$ for the leader omni-directional mobile robot consisted of the given four security points is as follows

$$\mathbf{D}_{safe} = \mathbf{Co}\{d_1, d_2, d_3, d_4\}$$

where $d_i = -[cos(2\pi i/4), sin(2\pi i/4)]^{\mathrm{T}}$, $i = 1, 2, 3, 4$; the maximum iteration duration $T_{max} = 30$, iteration interval $h = 0.01$ and threshold $tol = 0.1$; the initial centroid coordinates of the leader and follower omni-directional mobile robots are $[1, 0]^{\mathrm{T}}$ and $[5, 1]^{\mathrm{T}}$, respectively; $m = 2$ is the mass of omni-directional mobile robots; the initial tracking

Algorithm 1 Leader-follower tracking control of two omni-directional mobile robots

Pre-specify maximum iteration duration $T_{\max}$ and iteration interval h. Determine the initial coordinates of the centroid for the leader and follower omnidirectional mobile robots as r_1 and r_2, security points d_i, initial tracking error e, the disturbance γ and threshold tol. Set the parameter σ, initial matrices K_1, K_2 and λI in (14) and (15), and scalars a and b in (31). Specify the initial matrices K_0, G_0, P, $G_{\mathbf{d}}$. At time $t \geq 0$, perform the following steps:

(1) For the linear controller, calculate u as (16); for homogeneous controller, calculate u as (30).

(2) Implement the control input u to system (6), and update the tracking error e.

(3) If $\|e\|_2 \geq tol$, then return to step (1) and update the matrices K_1, K_2, and λI by utilizing (14) and (15); else if $\|e\|_2 < tol$, the matrices K_1, K_2, and λI remain unchanged. Then, return to step (1).

error $e = [3, 1]^{\mathrm{T}}$ and the initial system state $x = [e, \dot{e}]^{\mathrm{T}} = [3, 1, 0, 0]^{\mathrm{T}}$. The matrices A and B are as follows

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

The disturbance γ is described as

$$\gamma = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} [r_1; \dot{r}_1] - [sin(t), cos(t)]^{\mathrm{T}}$$

where $u_1 = [sin(t), cos(t)]^{\mathrm{T}}, Q_2(r_2, \dot{r}_2) = 0$ and $Q_1(r_1, \dot{r}_1) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} [r_1; \dot{r}_1]$. By utilizing (14) and (15) and the parameter $\sigma = 2$, the initial matrices K_1, K_2, λI and K_{lin} at time $t = 0$ are detailed as

$$K_1 = \begin{bmatrix} -0.5 & 0 \\ 0 & -0.5 \end{bmatrix}, K_2 = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}, \lambda I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$K_{lin} = \begin{bmatrix} -0.5 & 0 & -2 & 0 \\ 0 & -0.5 & 0 & -2 \end{bmatrix}$$

If u is selected as the linear controller (16), then $u = K_{lin}x$; if u is chosen as the homogeneous controller (30), then $u = K_0 x + u'_{hom}$ where $\mu = -1$ and $\|x\|_{\mathbf{d}} = 2.6918$ and $K_0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$; the parameters $a = 1$ and $b = 0.1$. The matrices $G_{\mathbf{d}} = \mathrm{diag}\{2, 2, 1, 1\}$, and G_0, P are represented as

$$G_0 = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, P = \begin{bmatrix} 5.25 & 0 & 4 & 0 \\ 0 & 5.25 & 0 & 4 \\ 4 & 0 & 5 & 0 \\ 0 & 4 & 0 & 5 \end{bmatrix}$$

The simulation results are shown in Fig.2–8. Fig.2 and Fig.3 illustrate the variations of the x-coordinates and y-coordinates of centroid for both the follower and leader omni-directional mobile robots with homogeneous control and linear control, respectively. The results illustrate that the x-coordinate and y-coordinate of centroid for follower omni-directional mobile robot are able to track the x-coordinate and y-coordinate of centroid for leader omni-directional mobile robot with homogeneous control. Fig.4 and Fig.5 indicate the trajectories of centroid for both the follower and leader omni-directional mobile robots with homogeneous control and linear control, respectively. It is shown that the homogeneous controller ensures that follower omni-directional mobile robot is capable of tracking the leader omni-directional mobile robot and performing the same motions with leader omni-directional mobile robot. In Figs 6–8, the control inputs, the tracking errors and their norm are given. It is indicated in Fig.6 that at the initial time, the homogeneous control has a larger control input compared with linear controller. However, Fig.7 shows that the tracking errors e_x and e_y in the related x and y coordinates with homogeneous control can achieve to 0 in finite-time. From the norm of the tracking error in Fig.8, it is shown that the tracking error for the homogeneous control can be stabilized to 0 in finite-time. The simulation results demonstrate that the homogeneous control can achieve finite-time stable and has a strong robustness compared with linear control.

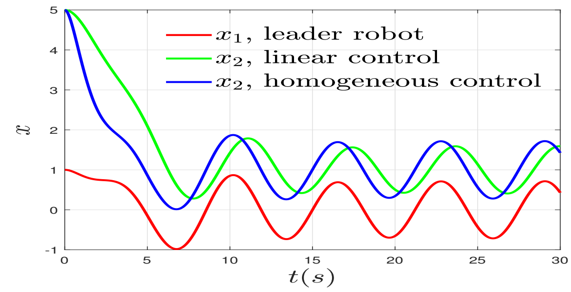

Fig. 2: The x-coordinates of centroid

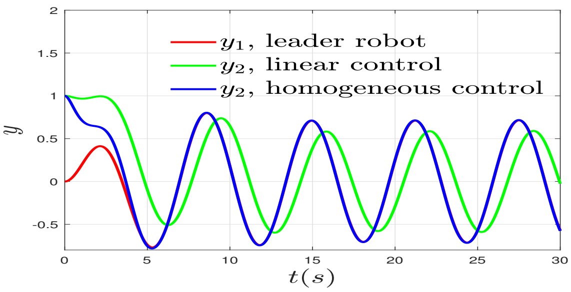

Fig. 3: The y-coordinates of centroid

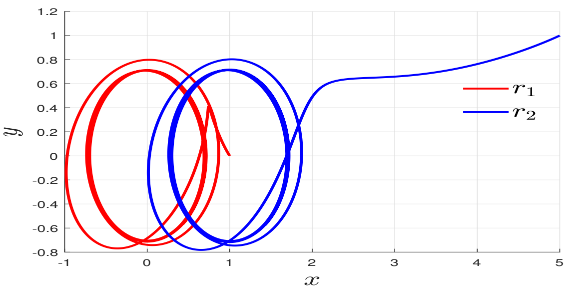

Fig. 4: The trajectories of centroid with homogeneous control

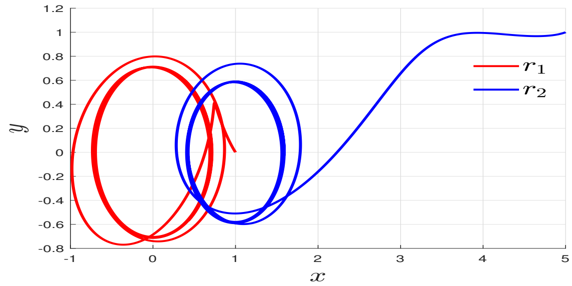

Fig. 5: The trajectories of centroid with linear control

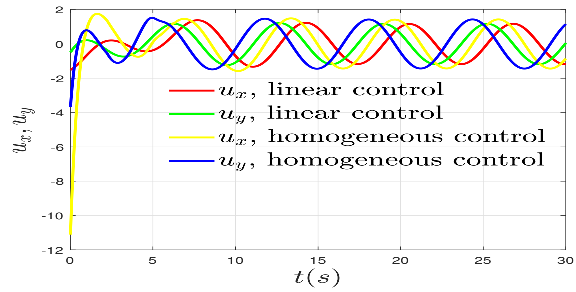

Fig. 6: The control inputs u

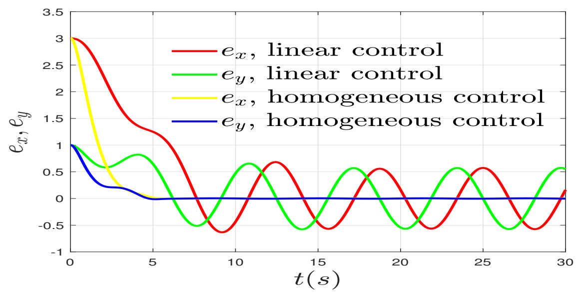

Fig. 7: The tracking error e

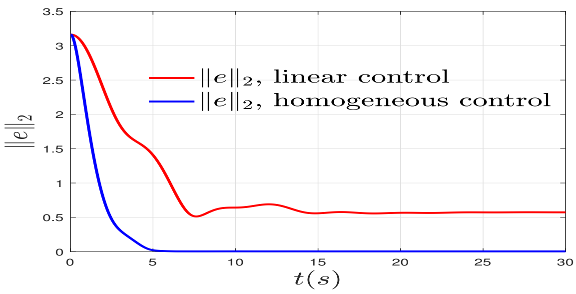

Fig. 8: Norm of the tracking error e

7 Conclusion

In this paper, a linear controller is firstly proposed for the leader-follower tracking control problem in two omnidirectional mobile robots and the asymptotical stability is guaranteed. In order to achieve finite-time control, a homogenous controller is developed based on the linear controller. Compared with the linear controller with asymptotical convergence, the finite-time stabilization implemented by using the homogeneous controller has faster convergent rate and better robustness, which is illustrated by the simulation results. Furthermore, the finite-time convergent rate of the homogeneous controller can be simply tuned by properly selection of a homogeneity degree.

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